

02393 Programming in C++

Module 1:

An overview of C++

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Lecture plan

Module	Date	Title	Textbook chapters
0 and 1	03/09	Welcome & C++ Overview	1
2	10/09	Basics of C++ and its Data Types	1.5, 2.1, 2.3 - 2.5, 11.1, 11.3
3	17/09	Memory management	12.1, 11.2, 11.3
4	24/09	LAB DAY	N/A
5	01/10	Libraries and Interfaces	2.2, 2.6 - 2.8, 3.1 - 3.3, 4.1 - 4.3
6	08/10	Classes and Objects	5.1, 6.1, 11.2, 12.1, 12.4, 12.7
Autumn break			
7	22/10	Templates	5.1, 14.2
8	29/10	LAB DAY + Guest lecture	N/A
9	05/11	Inheritance	19.1 - 19.3
10	12/11	Recursive Programming	7
11	19/11	Linked Lists	12.2, 12.3
12	26/11	Trees	16.1 - 16.4
13	03/12	LAB DAY	N/A

Outline

- What is C++?
- A brief look at C++ programs
- Recommended C++ compiler and editor
- Lab time

What is C++

- C++ is a general-purpose programming language that extends C
 - Created by Bjarne Stroustrup (beginnings: 1979)
 - Adopted by the industry (circa 1990 onwards)
 - Now regulated by ISO C++ Standard Committee (C++98, C++03, . . . , C++20, . . .)
- Unique combinations of
 - High level features (e.g., object oriented and generic programming)
 - low-level features (manual memory management, memory pointers)

C++ compared to other programming languages

- C++ vs Python, Matlab, etc.
 - Statically typed, compiled, manual memory handling (no garbage collection)
- C++ vs. Java, C#, Visual Basic, etc.
 - Manual memory handling (no garbage collection)
- C++ vs C
 - Object-oriented, more advanced types
- C++ vs Rust
 - Object-oriented, less advanced types, more mature, bigger ecosystem

An overview of C++ programs

- The structure of a C++ program
 - #include directives, the main function, user-defined functions
- Simple input/output
 - cin, cout
- Variables, values and types
 - string, int, double, float, bool
- Expressions
 - some numeric and boolean operators and math functions
- Statements
 - if, while, for

A very simple C++ program

```
1  int main()  
2  {  
3      int myVar = 3;  
4  
5      if (myVar < 2) {  
6          return 0;  
7      } else {  
8          return 1;  
9      }  
10 }
```

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- main() function:
 - Must always exist
 - Will be invoked when the program is run
 - Will return an integer value
 - By convention, if the return value is equal to 0, it indicates that the program ran successfully
 - If not equal to 0, it indicates that an error occurred

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```

- Variable declaration:
 - An integer value named myVar is declared and initialized to 3
 - It is possible to declare variables without initializing them
 - In this case, the variable will contain a random value
 - Uninitialized variables can be accessed later on in the program
 - If not properly initialized, this can lead to nasty bugs
 - It is a good practice to initialize variables either while they are declared, or immediately after

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5      if (myVar < 2) {  
6          return 0;  
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9      }  
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```

- If/else statement:
 - Checks if myVar is lower than 2
 - If so, it executes the code inside the if part of the block
 - If not, it executes the code inside the else part of the block

A very simple C++ program

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```

- Return statement:
 - Terminates the execution of main() by passing the specified value to the caller
 - In this case, the value is a constant, but it can also be a variable
 - For the main block, the caller is the operating system that runs our program

A classical beginner's program: hello world!

- The cout object represents the standard output (data that is shown as the text output of a program)
 - It can be used for our program to output messages

```
1  #include <iostream>
2
3  using namespace std;
4
5  int main()
6  {
7
8      cout << "Hello World!" << endl;
9
10     return 0;
11 }
```

A classical beginner's program: hello world!

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 - It can be used for our program to output messages
- To output a message, the “Hello World!” string is passed to cout via the << operator

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5  int main()
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7      cout << "Hello World!" << endl;
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9      return 0;
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```

- The cout object represents the standard output (data that is shown as the text output of a program)
 - It can be used for our program to write messages on screen
- To output a message, the “Hello World!” string is passed to cout via the << operator
- To write subsequent output on a new line, the endl object is passed to cout after the Hello World!” string
 - If endl is not passed, any subsequent output would be written on the same line!

A classical beginner's program: hello world!

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```

- The cout and endl objects are not part of the C++ standard constructs
 - Instead, they are part of the iostream library
- The #include statement tells the compiler that our program needs the iostream library to work

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```

- Library functions and objects with the same name may exist in different libraries
- To disambiguate them, namespaces are used
 - They work similarly to packages in other programming languages
- For the `iostream` library, `cout` and `endl` live inside the `std` namespace
- To tell the compiler in which namespace to look for them, one should use `std::cout`
- To save some typing, the `using namespace std` statement tells the compiler to use `std`

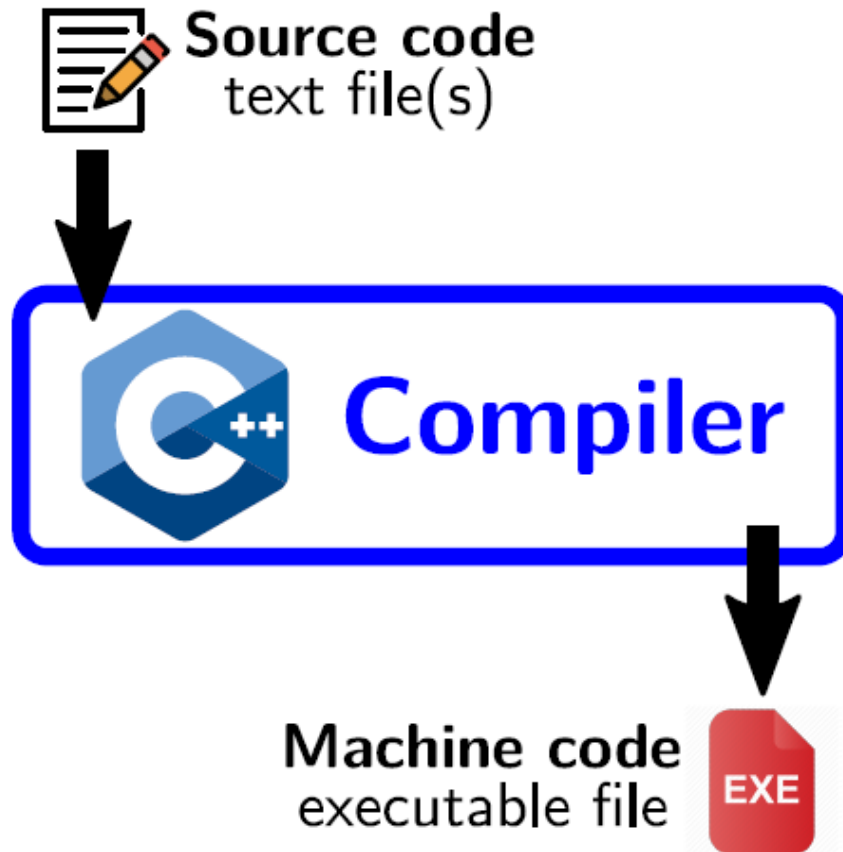
Live code example: a simple calculator

- I will now show C++ in action, by using an online C++ editor and compiler: <http://cpp.sh>
 - Try to follow along
 - If something is unclear, please raise your hand
 - For the next lectures, you will need to install a C++ compiler on your computer (more on this later)

Break

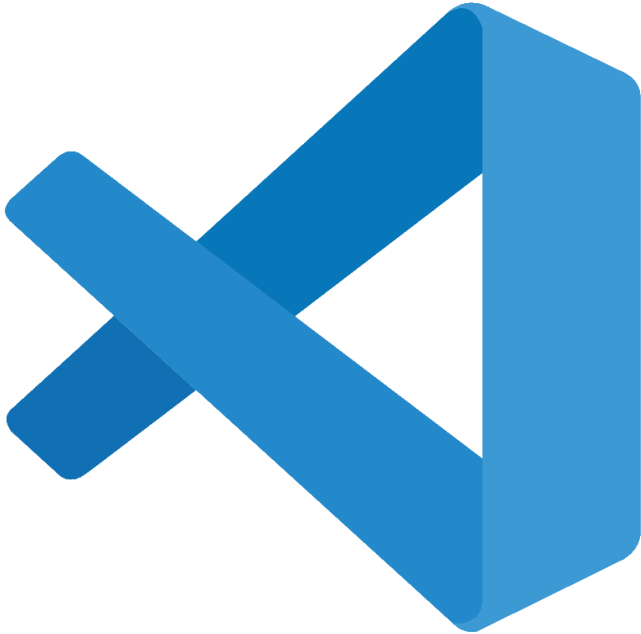
- We resume at 19:00

The compile process



- A C++ compiler is a program that:
 - reads source code from one or more text files
 - generates an executable file containing machine code
- You can use any text editor to write the source code
 - ...but maybe you want syntax highlighting at least
- There are many C++ compilers with various differences:
 - different compatibility with C++ language standards
 - different usage options
 - different error messages

Recommended for this course



- Microsoft VS Code (not Visual Studio) + GNU Compiler Collection (which includes the C++ compiler g++)
 - Both platforms are open source and exists for most desktop platforms
 - g++ is standard compliant and used on Autolab
- See instructions on Autolab on how to configure your environment

Lab time & homework

- Install Visual Studio Code and GCC (g++) on your computer
 - Follow the instructions on DTU Learn
 - If you have trouble, ask for help
 - Mac OS users: you may have a compiler called g++ that is not GCC. It should be OK. . .
- Try the first weekly assignment on your computer and on Autolab
 - The assignment description is available on DTU Learn
 - Goal: check your C++ installation and familiarize with Autolab
 - Don't worry if you don't get all exercises right :-)

DTU

